The hyperelliptic head-to-head as ecbench records

research/ic_acct3_hec_ecbench_20261021 · session 2026-10-21 · report 2026-10-26 · class: accounting port

Question

research/hec_bitop_20261020 priced hyperelliptic index calculus and its Pollard rho reference in bitop/v1 through a standalone example, with no sealed records and no replay. This item ports both arms into ecbench as the methods hec.ic_fb_walk and hec.rho_full_width (commit 8c37d9a6e) and asks one question, frozen in PROTOCOL.md before the session ran:

Run through ecbench on the study's genus-2, $p = 41$ row ($N = 1321$, $k = 447$), with the study's instance, target and seeds, do the records' bitop/v1 totals equal the study's bitops_per_run, run for run, for both arms?

The decision is exact integer equality, computed by the Rust example examples/hec_ecbench_equivalence.rs. This is an accounting port. It changes no IC cost and is not an improvement of the index-calculus baseline.

Session

- **Spec** specs/g2-p41.json (ECS1hf41f08dfbb2f) → session sessions/g2-p41 (ECBS1he77b325d1865). Built at 1961d1878 with a dirty tree (the port under review; git_dirty: true in session.json), binary sha256 055fa754….
- **Arms.** 5 IC arms (fixed seeds 20260916–20260920; branches 64, d_step_in 4, extra_relations 8) and 40 rho arms (fixed seeds 0xC0FFEE + 0…39). One round, no warm-up, --cpus none, RAYON_NUM_THREADS=1. Isolation L0: no wall-clock figure is used.
- **Runs.** 45 requested, 45 records, 45 verified ($[k]D_1 = Q$ checked by the runner; neither method sees the planted scalar).
- **Audit.** ecbench verify --replay 2: ok, 0 problems; replays of seq 0 (IC) and seq 22 (rho) both identical (sessions/g2-p41.audit.json).
- **Pricing.** ecbench work --counted: all 45 records complete, nothing unpriced, nothing known-uncounted (analysis/work.jsonl).

Result: equivalent

All 45 per-run totals and all 12 per-phase sums (6 phases × 2 arms) are equal to the committed study counts (analysis/equivalence.md).

arm	runs	equal	ecbench Σ bitop/v1	study Σ bitop/v1	mean per run
hec.ic_fb_walk	5	5	117,422,628	117,422,628	23,484,525.6
hec.rho_full_width	40	40	4,535,211,568	4,535,211,568	113,380,289.2

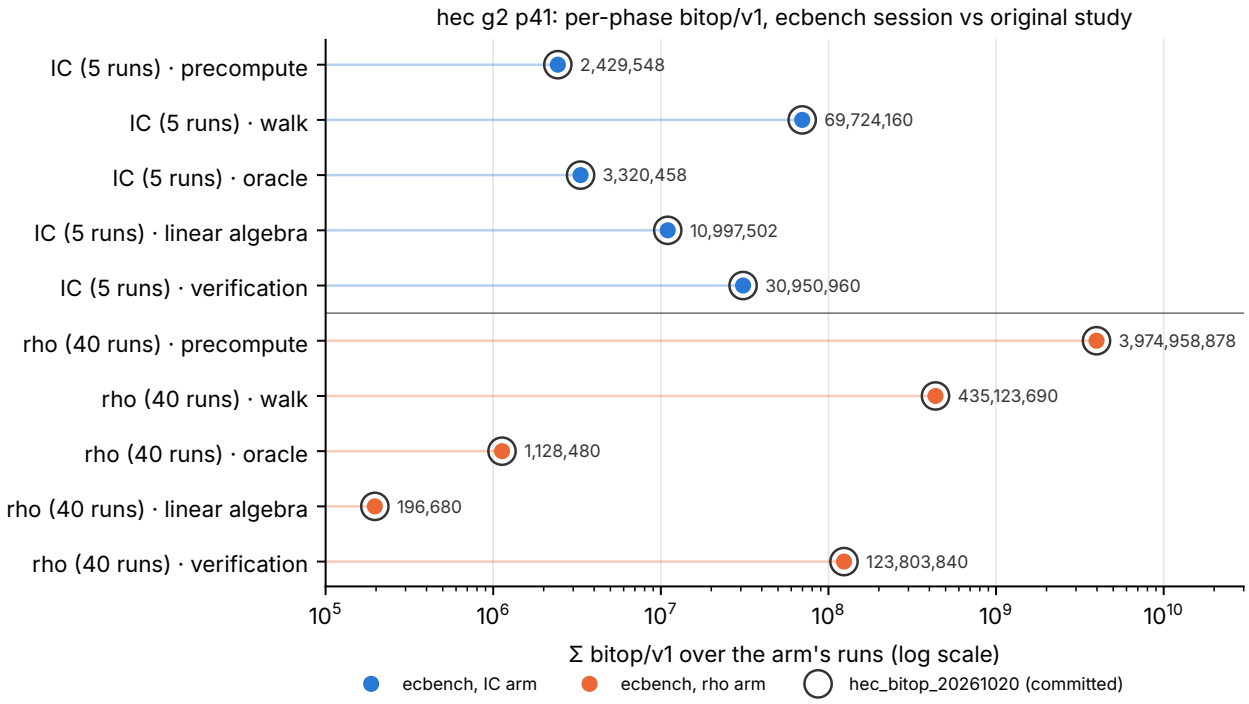


Figure 1: Per-phase `bitop/v1` summed over each arm's runs, session `ECBS1he77b325d1865` (filled) against `research/hec_bitop_20261020` (ring), from `analysis/equivalence.md`. Log scale. The `unattributed` phase is 0 in both, for both arms, and is not drawn. Counts are exact; there is no sampling interval to show.

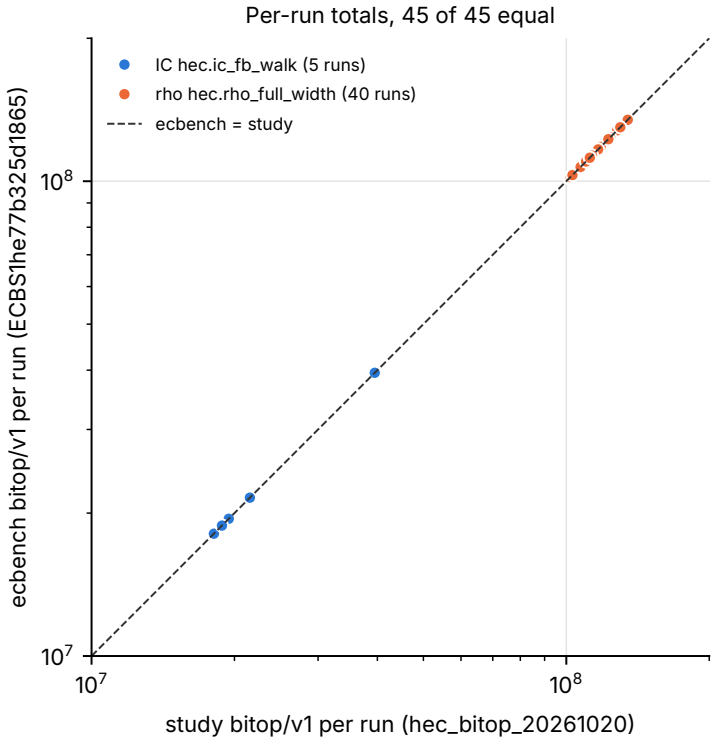


Figure 2: Per-run `bitop/v1` totals, ecbench (y) against the study (x), all 45 runs (5 IC, 40 rho). Every point lies on $y = x$ by exact integer equality.

Position against rho. The IC/rho ratio carried over is the study's 0.207 at this row (mean `bitop/v1` per run, 23.5 M against 113.4 M), so this row already sits below its matched full-width rho, with the study's disclosed coverage limits (`research/hec_bitop_20261020/PROTOCOL.md`). It is the study's figure reproduced through ecbench, not a new measurement or an improvement. The rho arm's sum is dominated by its `precompute` phase (3.97e9 of 4.54e9 bitops); the IC arm's by `walk` (6.97e7 of 1.17e8). These are the study's recorded splits.

Limitations

- **One row.** Genus 3 and 4 and the other primes are not run here; one size gives no slope, so no bound is fitted.
- **GAE figure.** The records' GAE count holds only the Jacobian operations the algorithms count, and they name `jacobian_field_ring_and_word_work_outside_group_operations` as unpriced, so `S` is a lower bound. The `bitop/v1` work block is the complete figure.
- **Floor.** `work`'s `floor_bitops` uses an elliptic affine addition's price; it is not a calibrated floor for a Jacobian.
- **Conventions** are the study's: one word operation per RNG draw, memory movement uncounted, validation inside the total.
- **Validation gate.** There is no `VALIDATION.json` (no replay certificate or validator report), so nothing here goes to the IC leaderboard or `benchmarks/README.md`; it is an accounting port, not a performance result.
- **No instruction audit** (`ecbench audit-ir`) was run for these records.

Evidence

`PROTOCOL.md`, `specs/g2-p41.json`, `sessions/g2-p41/` (records, plan, host, session), `sessions/g2-p41.audit.json`, `analysis/work.jsonl`, `analysis/equivalence.md` (`examples/hec_ecbench_equivalence.rs`), `figures/plots.py` (parses `equivalence.md`; plots recorded values only).